Dheirya Tyagi

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Education

Georgia Institute of Technology — Atlanta, GA
Bachelor of Science in Computer Engineering — *Expected May 2029*

Experience

Founder & Lead Developer — Nov. 2024 – Present
Simple Explanations — *Remote*

- Grew a STEM education channel to 7.5M+ students and 35K+ followers across YouTube and TikTok by producing 370+ videos and applying audience retention analytics to refine scripting, pacing, and topic selection.
- Recognized a gap in free, accessible study resources and founded a platform combining a curated link database, 125+ crowdsourced study guides with 50K+ views, and adaptive quizzes using FastAPI and PostgreSQL.

RoboSub Subsystem Builder — Aug. 2026 – Present
Georgia Tech Marine Robotics Group — *Atlanta, GA*

- Contribute to the electrical build of an autonomous underwater vehicle rated to 10m depth for the RoboSub competition, coordinating with a 5-person hardware subteam across frame design, waterproofing, and power.
- Wiring and integrating 8 thrusters with electronic speed controllers and a custom power distribution board, working directly with the software and autonomy subteams to align hardware with control-code configuration and testing.

Research Intern & Team Leader — May 2025 – Feb. 2026
New Jersey Cybersecurity & Communications Integration Cell — *Ewing, NJ*

- Strengthened the state's threat assessment strategy by identifying quantum computing as an emerging threat to network cybersecurity, researched its evolving impact, and presented findings to state cybersecurity professionals.
- Extended cybersecurity outreach to 4K+ students and vulnerable audiences across New Jersey by leading an 8-person team to design instructional CTF challenges, hands-on exercises, and a newsletter.

Projects

Braitenberg Light-Seeking Vehicle | *PCB Design, PCB Fabrication, Analog Circuits, Soldering* — Fall 2026

- Built an autonomous robot that tracks and drives toward light sources using a fully analog control circuit, with zero firmware or a microcontroller.
- Implemented proportional steering by cross-coupling two photoresistors to opposite motors through MOSFET drivers, with flyback diodes protecting the circuit from motor voltage spikes.
- Laid out and fabricated a two-layer PCB in-house via LPKF laser etching and ProtoMat drilling, then soldered, continuity-tested, and integrated it with custom 3D-printed PETG mounts.

Bernbox Arcade Machine | *Raspberry Pi, Embedded Linux, Woodworking* — Spring 2026

- Engineered a six-foot, two-player arcade cabinet around a Raspberry Pi 3B+ running a Linux-based OS. Wired and soldered HAPP-style buttons and joysticks through USB encoder boards.
- Optimized encoder-to-emulator input mapping, running 50+ games across 10+ platforms on low-power hardware.
- Deployed the cabinet for daily public use in the school library, leading team through a two-year agile build.

Electromagnetic Seismograph | *Arduino, Python, Serial Communication* — Spring 2026

- Designed a seismograph that converts ground vibrations into voltage via a magnet-and-coil induction sensor, sampling at 20Hz with ∼1mV resolution using the internal 1.1V ADC reference.
- Developed an Arduino-Python serial pipeline that streams live data to a 25-second rolling plot and drives a 60-LED readout that scales color and intensity with vibration magnitude.

Optic Prime | *Arduino, C++, Ultrasonic Sensors* — Fall 2025

- Prototyped an Arduino navigational aid for visually-impaired people using ultrasonic sensors with graded obstacle feedback between 0.5m–3m, providing real-time haptic response scaling from 80–600ms.
- Programmed C++ firmware to filter noisy sensor readings, tuning thresholds via field testing with 10+ volunteers.
- Won 1st overall out of 30+ teams at BRHacks medical-themed hackathon, earning $1000 in prize funding.

Technical Skills

Languages: Python, Java, C++, SQL, MATLAB, HTML/CSS/JavaScript
Hardware: Arduino, ESP32, Raspberry Pi, PCB Design, PCB Fabrication, Motor Control, Soldering, CAD
Software & Tools: Git, Linux, FastAPI, Django, ROS, OpenCV, MediaPipe, PostgreSQL